

A Mathematical Model of N-Linked Glycosylation

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Abstract: Metabolic engineering of N-linked oligosaccharide biosynthesis to produce novel glycoforms or glycoform distributions of a recombinant glycoprotein can potentially lead to an improved therapeutic performance of the glycoprotein product. A mathematical model for the initial stages of this process, up to the first galactosylation of an oligosaccharide, was previously developed by Umana and Bailey (1997) (UB1997). Building on this work, an extended model is developed to include further galactosylation, fucosylation, extension of antennae by N-acetylglucosamine repeats, and sialylation. This allows many more structural features to be predicted. A number of simplifying assumptions are also relaxed to incorporate more variables for the control of glycoforms. The full model generates 7565 oligosaccharide structures in a network of 22,871 reactions. Methods for solving the model for the complete product distribution and adjusting the parameters to match experimental data are also developed. A basal set of kinetic parameters for the enzyme-catalyzed reactions acting on free oligosaccharide substrates is obtained from the previous model and existing literature. Enzyme activities are adjusted to match experimental glycoform distributions for Chinese Hamster Ovary (CHO). The model is then used to predict the effect of increasing expression of a target glycoprotein on the product glycoform distribution and evaluate appropriate metabolic engineering strategies to return the glycoform profile to its original distribution pattern. This model may find significant utility in the future to predict glycosylation patterns and direct glycoengineering projects to optimize glycoform distributions.

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INTRODUCTION

Many valuable therapeutics are produced as recombinant glycoproteins that include oligosaccharides attached and modified during post-translational glycosylation. These recombinant proteins are generated in a number of expression systems including mammalian cell lines, insect cells,

yeast, and other hosts. While the recombinant glycoproteins generated by these expression systems often have the same polypeptide backbones as the human versions, the oligosaccharide structures can vary significantly (Betenbaugh et al., 2004). What is especially important is that the structural nature of these oligosaccharides can influence key properties of the pharmaceuticals, including biological activity, clearance rate, immunogenicity, solubility, and stability against proteolysis. (Cumming, 1991; Jenkins and Parekh, 1996). For example, the presence of sialic acid on these oligosaccharides and the degree of branching can be especially critical to the *in vivo* activity and hence the “quality” of many glycoprotein therapeutics such as EPO (Lemke and Muller, 1989; MacDougall, 2002). In addition, the sialic acid content and the degree of oligosaccharide branching can also be highly sensitive to protein expression rates and vary with the expression of target proteins (Santell et al., 1999). It has also been suggested that particular glycoforms of proteins could be susceptible to the misfolding and aggregation of the proteins into amyloid plaques of prion diseases and Alzheimers (Nowak et al., 1998). Indeed the great variety of glycoforms may have some value in reducing the probability of identical glycoforms meeting each other to form aggregation nuclei. Improved quantitative understanding of the glycosylation process could lead to new treatment options for these important diseases.

Various methods can be used to manipulate the glycosylation process in order to modify oligosaccharide structure. For example, the cellular environment has been shown to affect glycoform structure (Baker et al., 2001; Goochee et al., 1991; Sasai and Ikeda, 2002). In addition, heterologous glycosylation enzymes have been incorporated into cell lines in order to alter the oligosaccharide structures generated (Lee et al., 1989; Minch et al., 1995). Using such manipulations it may be possible to optimize the production system in order to obtain desired glycoform profiles or to produce novel structures with higher therapeutic efficiency. But the effects of these manipulations on the final glycoprotein structural pattern can be difficult to predict because of the complexity of the glycosylation process. A quantitative model of glycosylation could therefore be very useful in guiding experiments on cell culture or metabolic engineering manipulations.

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Umana and Bailey (1997) made a significant contribution to this goal with a mathematical model of N-linked glycosylation (UB1997 model). This model included the effects of enzyme distribution among four cellular compartments and followed the initial glycan processing steps including generic branching of the *N*-glycans. The model applied literature kinetic parameters based on Chinese Hamster Ovary (CHO) cell data to reach worthwhile qualitative conclusions about effects of enzyme activities on glycan structure.

The current paper expands the UB1997 model in order to allow it to predict a wider range of oligosaccharide structures encountered in mammalian cell expression systems. In addition, the model can now be applied to a larger range of experimental conditions that might be encountered in a cell culture environment. The scope has been expanded by inclusion of additional processing steps including the extension of branches by the LacNAc disaccharides and the capping of *N*-glycan branches with sialic acid. The effects of donor substrate concentrations required for the transferases are also included in the kinetic equations in order to consider a wider range of intracellular conditions. These extensions provide more structural details and will enable a more robust analysis of the effects of environmental and genetic manipulations on structural changes. The extensions increase the number of structures and reactions considerably from the 33 structures and 33 reactions of the UB model to thousands of structures and tens of thousands of reactions. To handle this expansion the structures and reaction networks are generated automatically from a system of reaction rules. Methods used to do this are reminiscent of those used for similarly complex models in other fields, including combustion and petroleum processing (Grenda et al., 2003; Quann and Jaffe, 1996; Quann and Krambeck, 1991). Mathematical methods for adjusting model parameters to match experimental glycan structure distributions and for optimizing model parameters to achieve desired distributions are also included.

An additional complication of glycosylation is that the conformation of the polypeptide chain in the neighborhood of the glycan attachment site affects the distribution of structures at that site (Kawasaki et al., 2000). Because of this, each possible glycosylation site on a single glycoprotein can have a different set of kinetic parameters. In many cases, the overall glycan distribution for a glycoprotein is measured rather than the distribution at each individual site. This study assumes that all the sites have the same kinetics, which could be interpreted as modeling only one of the several sites on a given protein or as modeling the average kinetics of all the sites. The latter assumption is used to compare the model with experimentally measured glycan structure distributions.

This paper describes how the current model has been developed in order to provide agreement with experimental glycan distributions for CHO cells in culture. Next, it is shown how the model can be used to predict the effects of changes in the extracellular environment due to media changes or alterations in the expression levels of target proteins. Finally, the model is used to predict how the expression

levels of particular oligosaccharide modifying enzymes can be modified in order to optimize glycosylation processing and obtain a desirable target oligosaccharide distribution pattern.

GLYCOSYLATION PROCESS

Material Transport

Glycosylation of secreted and membrane glycoproteins takes place in the endoplasmic reticulum (ER) and Golgi compartments of the cell. These include the *cis*-Golgi network (CGN), the *cis*-, medial, and *trans*-Golgi cisternae (CGC, MGC, and TGC) and the *trans*-Golgi network (TGN). The glycoproteins migrate through these membrane-enclosed compartments in sequence as part of a budding process and encounter different concentrations of the various enzymes involved, which are each localized in particular compartments. Analysis of these processes has indicated that the apparatus can be approximated mathematically as a sequence of well-mixed reactors with bulk transport of proteins and other mobile compounds moving from each compartment to the next in sequence (Monica et al., 1997; Umana and Bailey, 1997). This assumption is also used in the current model. Also, as in the UB1997 model, the CGC is set as the first compartment of the model, since only one or a few different glycan structures enter this zone.

Processing in the Golgi network involves removal of mannose groups along with addition of various sugar residues onto the growing glycans during the glycosylation process. Removal of the mannose groups is facilitated by cellular mannosidases while the addition of different sugar residues is driven by specific glycosyltransferases. The glycosyltransferases transfer the particular sugar from sugar-nucleotide donor molecules such as UDP-GlcNAc onto the acceptor oligosaccharide chain. These sugar-nucleotides are actively transported from the cytoplasm into the compartments via an antiporter mechanism with the corresponding nucleotide end product (in this case UMP) as the antiporter. As a result, the concentrations of the sugar-nucleotide species are much higher within the compartments than in the cytoplasm (Perez and Hirschberg, 1985; Waldman and Rudnick, 1990). While these active transport processes could be included in the model the present version assumes fixed concentrations of the donors within the compartments.

The previous model did not require any knowledge of the donor substrate concentrations and instead assumed that the enzymes were saturated with these cosubstrates. In contrast it has been shown that the donor concentrations do affect the glycan structures. In one study it was shown that the level of UDP-GlcNAc significantly affects branching of the oligosaccharide chains. (Baker et al., 2001). K_m values for the donors can be quite high, especially for GlcNAc transferases IV and V (Sasai and Ikeda, 2002; Taniguchi et al., 1989). Based on these studies, the effects of donor concentration have been included in the current model.

N-Glycan Structures

N-glycans are linked to the polypeptide backbone of glycoproteins through asparagine (Asn) residues that are part of an Asn-X-Ser/Thr sequence. Glycoproteins often have several such potential glycosylation sites which may or may not be occupied by an N-glycan. The attached N-glycans share the common core structure shown in Figure 1a. A wide variety of oligosaccharides can attach to this core as described by Schachter et al. (1989).

This model expands considerably the variety of structures considered compared to the UB1997 effort but does not yet include all known structures, only those generated by a subset of the known enzymes. A list of the enzymes considered in the present model is listed in Table I. It includes all the enzymes commonly encountered during N-glycosylation of glycoproteins in CHO cells. One of them, GnTIII, which adds a bisecting GlcNAc, has not been observed in wild-type CHO but has been observed in other mammalian cell lines and the heterologous gene has been engineered into recombinant CHO cells (Stanley and Campbell, 1984; Sburlati et al., 1998; Umana et al., 1999).

The initial model input structure shown in Figure 1b results from the original addition of an oligosaccharide precursor in the ER to one of the glycosylation sites followed by reactions in the ER and CGN that remove three Glucose residues and in some cases a Mannose to give a mixture of M_9 (shown) and M_8 structures entering the CGC, which is the first compartment of the model. The M_9 and M_8 mixture is the starting point of the model. Following UB1997 this is assumed this to be a 50:50 mixture of the two.

A typical N-linked oligosaccharide (N-glycan), as shown in Figure 1c, generated in the N-glycosylation processing pathway can contain up to four branches, or antennae, each containing a variety of linked monosaccharides. An additional GlcNAc group, known as a bisecting GlcNAc, can also be linked directly to the core mannose group in some mammalian cell lines. A fucose group can also attach to the innermost GlcNAc of the structure. Shown in Figure 2 are the initial stages of the reaction network. These are all the structures that result from the action of just the enzymes ManI, ManII, and GnT1–GnTV on the initial M_9 and M_8 mixture. The initial M_9 or M_8 glycan that enters the CGC is first acted on by Golgi $\alpha(1,2)$ -mannosidase I (ManI) which removes the four $\alpha(1,2)$ -linked mannose residues one at a time, converting M_9 to M_8 , M_8 to M_7 and so on, until the M_5 structure is reached. The mannose residues are removed in a specific order, so that there are only two M_8 structures, and only one each of M_7 and M_6 (Tabas and Kornfeld, 1979). The model does not differentiate between the two M_8 structures.

The next step is addition of a GlcNAc residue to the $\alpha(1,3)$ -connected (lower) core mannose group through the action of GnTI. After the Gn M_5 structure is obtained, the network can branch in multiple directions. The Gn M_5 can be acted upon by ManII to remove a terminal $\alpha(1,3)$ -connected mannose group, by GnTIII to add a bisecting GlcNAc to the central core mannose group, or by GnTIV to add another GlcNAc group connected $\beta(1,4)$ to the lower core mannose group. These three alternatives are shown in the second column of Figure 2. As shown in subsequent columns of Figure 2, ManII can remove the remaining terminal $\alpha(1,6)$ -connected mannose group after the $\alpha(1,3)$ -connected terminal mannose

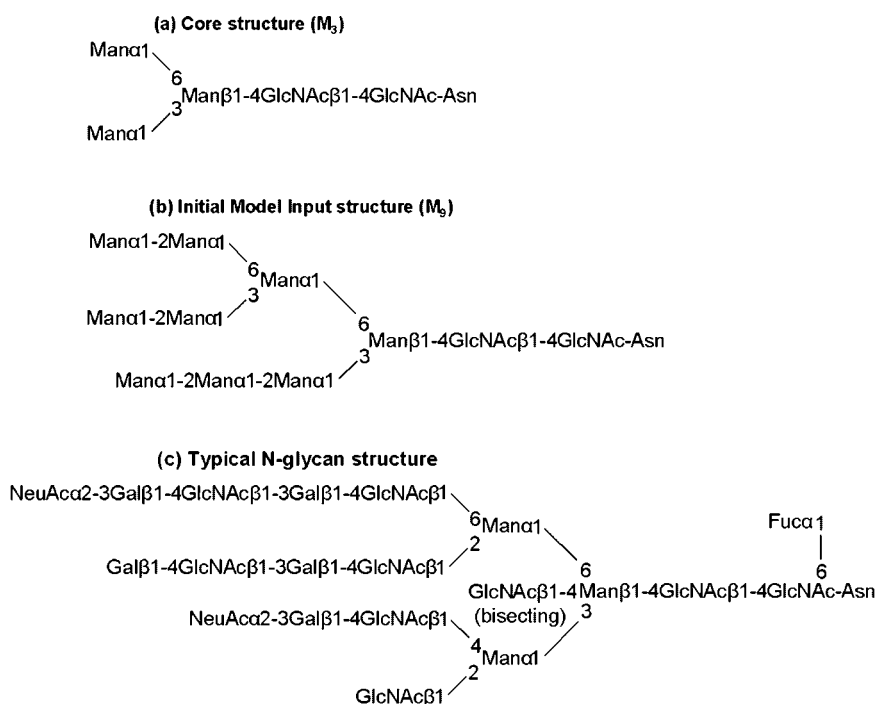


Figure 1. Glycan structures. Man = mannose, GlcNAc = N-acetylglucosamine, Fuc = fucose, Gal = galactose, NeuAc = N-acetylneuraminic acid (sialic acid).

Table I. Enzymes included in present model (for enzyme details see Schomburg et al., 2004 <http://www.brenda.uni-koeln.de/>).

Abbreviations	EC No.	Name
ManI	3.2.1.113	Mannosyl-oligosaccharide 1,2- α -mannosidase
ManII	3.2.1.114	Mannosyl-oligosaccharide 1,3-1,6- α -mannosidase
FucT	2.4.1.68	Glycoprotein 6- α -L-fucosyltransferase
GnTI	2.4.1.101	α -1,3-mannosyl-glycoprotein 2- β -N-acetylglucosaminyltransferase
GnTII	2.4.1.143	α -1,6-mannosyl-glycoprotein 2- β -N-acetylglucosaminyltransferase
GnTIII	2.4.1.144	α -1,4-mannosyl-glycoprotein 4- β -N-acetylglucosaminyltransferase
GnTIV	2.4.1.145	α -1,3-mannosyl-glycoprotein 4- β -N-acetylglucosaminyltransferase
GnTV	2.4.1.155	α -1,6-mannosyl-glycoprotein 6- β -N-acetylglucosaminyltransferase
GnTE	2.4.1.149	N-acetylglucosaminide β -1,3-N-acetylglucosaminyltransferase
GalT	2.4.1.38	β -N-acetylglucosaminylglycopeptide β -1,4-galactosyltransferase
SiaT	2.4.99.6	N-acetylglucosaminide α -2,3-sialyltransferase

group is gone. Once the upper core mannose group is exposed GnTII, followed by GnTV can act to add GlcNAc groups to the upper core, generating the tetraantennary structures observed experimentally in CHO and other mammalian cell lines.

Any of the structures with terminal GlcNAc groups (not bisecting) can add galactose groups by the action of GalT. The galactose groups, in turn, can accept sialic acid groups via SiaT or additional GlcNAc groups via GnTE. The latter addition provides a new terminal GlcNAc group that can again be galactosylated, thus generating a repeating Gal β 1-4GlcNAc (*n*-acetylglucosamine) group.

The best way to understand this network is through a set of reaction rules that specify the conditions under which each enzyme can act on a possible structure. To describe reaction rules and apply them in an automatic fashion to a large

number of reactions, a succinct method of describing structures that simplifies computer representation is needed. For this purpose each of the glycoforms is described by a series of nine numbers that define uniquely an oligomer structure. The definitions of the nine numbers are given in Tables II and III.

The first three numbers are the total number of mannose (Man), fucose (Fuc), and bisecting *N*-acetylglucosamine (Gnb) residues, respectively, in the *N*-glycan. The next four numbers, Br1, Br2, Br3, and Br4 describe each of the possible four structurally unique branches or antennae. If the branch is absent, its corresponding number is zero. Each of the four branches can have one of the six structures shown in Table III numbered 1 to 6. Branches 1 and 2 are attached to the α (1,6) linked (upper) mannose (Man) of the tri-mannosyl core oligosaccharide, and branches 3 and 4 are attached to the

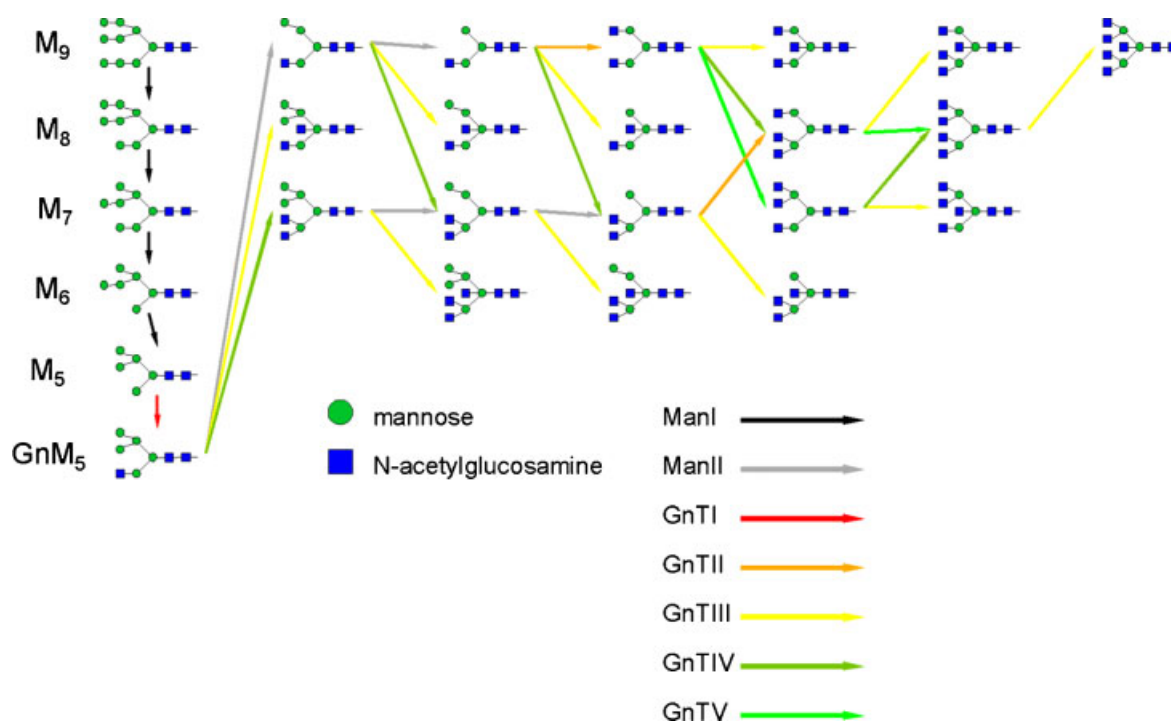


Figure 2. Initial buildup of oligosaccharide structures prior to galactose or fucose addition. [Color figure can be seen in the online version of this article, available at www.interscience.wiley.com.]

Table II. Glycoform description scheme.

Man	Number of mannose residues
Fuc	Number of fucose residues. Can be zero or one
Gnb	Number of bisecting GlcNAc residues
Br1	Extension level of branch 1. Can be zero to six
Br2	Extension level of branch 2. Can be zero to six
Br3	Extension level of branch 3. Can be zero to six
Br4	Extension level of branch 4. Can be zero to six
Gal	Number of galactose residues
Sia	Number of sialic acid (NeuAc) residues

$\alpha(1,3)$ linked (lower) Man of the core oligosaccharide. The Gn linkages of branches 1 through 4 are $\beta(1,6)$, $\beta(1,2)$, $\beta(1,4)$, and $\beta(1,2)$, respectively. Note that further extension of the branches can be accommodated by allowing extension levels higher than 6 but has not been considered here. The last two numbers are the number of galactose residues (Gal) and number of sialic acid residues (Sia). Note that these last two are redundant since the numbers can be calculated from the branch levels in Table III, but the numbers were included for convenience in setting up the reaction rules. As an example the structure of Figure 1c would be represented by the sequence of numbers (3 1 1 6 5 3 1 5 2), or Man-3 Fuc-1 Gnb-1 Br1-6 Br2-5 Br3-3 Br4-1 Gal-5 Sia-2.

It should be noted that the structures included in this scheme are not exhaustive of all species (Schachter et al., 1989). For example, in avian cells it is possible for a fifth antenna to attach to the upper core mannose via a $\beta(1,4)$ linkage through the action of GnTVI. Terminal sialic acid groups can also attach via an $\alpha(2,6)$ linkage. Galactose groups can attach directly to a terminal $\beta(1,4)$ linked galactose via an $\alpha(1,3)$ linkage. Fucose groups can add to terminal galactose in a $\beta(1,2)$ linkage. Galactosamine residues can also be present. Still, the current scheme encompasses most of the structures encountered in many mammalian cell lines commonly used for recombinant protein synthesis. Other alternative structures can be added readily to the framework as needed.

Reaction Network

The reaction network information can be described succinctly as a table that expresses a selection rule for the conditions a substrate must satisfy in order to undergo the

Table III. Extension levels.

Level	Branch structure
1	GlcNAc-
2	Gal β 1-4GlcNAc-
3	NeuAc α 2-3Gal β 1-4GlcNAc-
4	GlcNAc β 1-3Gal β 1-4GlcNAc-
5	Gal β 1-4GlcNAc β 1-3Gal β 1-4GlcNAc-
6	NeuAc α 2-3Gal β 1-4GlcNAc β 1-3Gal β 1-4GlcNAc-

particular reaction and how the structural numbers change as a result. For example, the 8 reaction rules that generate the 33 structures and 33 reactions of the UB model are listed in Table IV. The resulting network is shown in Figure 3.

Rule 1 for the enzyme ManI (Man > 5) states that this enzyme can act on the structures M_9 through M_6 , decreasing the number of mannose groups by one each time, as indicated by the -1 in the “Man” column of the table. Once the M_5 structure is reached rule 3 for GnTI (Man = 5 & Br4 = 0) can act to add a Gn group to branch 4. The 1 in the column “Br4” of the table indicates that the value of Br4 is increased from zero to one. This then allows rule 2 for ManII (Man > 3 & Br4 = 1 & Gnb = 0) to act to remove two more mannose groups. Note that the presence of a bisecting GlcNAc shuts down this enzyme. Rule 5 for GnTIII (Gnb = 0 & Br4 > 0 & Gal = 0) allows the addition of a bisecting GlcNAc whenever there is at least one terminal GlcNAc but no terminal galactose groups. Rules 5 and 6 for GnTIV and GnTV may be overly restrictive as discussed below, requiring that both branch 2 and branch 4 be present and that there be no galactose.

The UB1997 model is aimed at following just the main structure-forming steps in the process. As a simplification the UB1997 model assumes that once galactose adds to any branch all the GlcNAc transferases shut down so that the overall structures are determined at that point. Subsequent additions to the branches are not considered. Rather than separately considering the addition of galactose to each of the terminal GlcNAc groups in a structure, the UB1997 model lumps these separate reactions into a single reaction with rate parameters approximating the total rate of the individual reactions. Subsequent galactosylation is not considered. This is represented in Table IV by assuming that galactose only adds to branch 4, which will always be present if any branch is present.

Table IV. Reaction Rules for UB1997 model (33 structures, 33 reactions). The resulting reaction network is shown in Figure 3.

Index	Enzyme	Cosubstrate	Coproduct	Substrate rule	Man	Fuc	Gnb	Br1	Br2	Br3	Br4	Gal	Sia
1	ManI	water	mannose	Man > 5	-1								
2	ManII	water	mannose	Man > 3 & Br4 = 1 & Gnb = 0	-1								
3	GnTI	UDP-GlcNAc	UDP	Man = 5 & Br4 = 0							1		
4	GnTII	UDP-GlcNAc	UDP	Man = 3 & Br4 = 1 & Br2 = 0 & Gnb = 0				1					
5	GnTIII	UDP-GlcNAc	UDP	Gnb = 0 & Br4 > 0 & Gal = 0			1						
6	GnTIV	UDP-GlcNAc	UDP	Br4 > 0 & Br2 > 0 & Br3 = 0 & Gnb = 0 & Gal = 0					1				
7	GnTV	UDP-GlcNAc	UDP	Br2 > 0 & Br1 = 0 & Gnb = 0 & Gal = 0				1					
8	GalT	UDP-Gal	UDP	Br4 = 1							1	1	

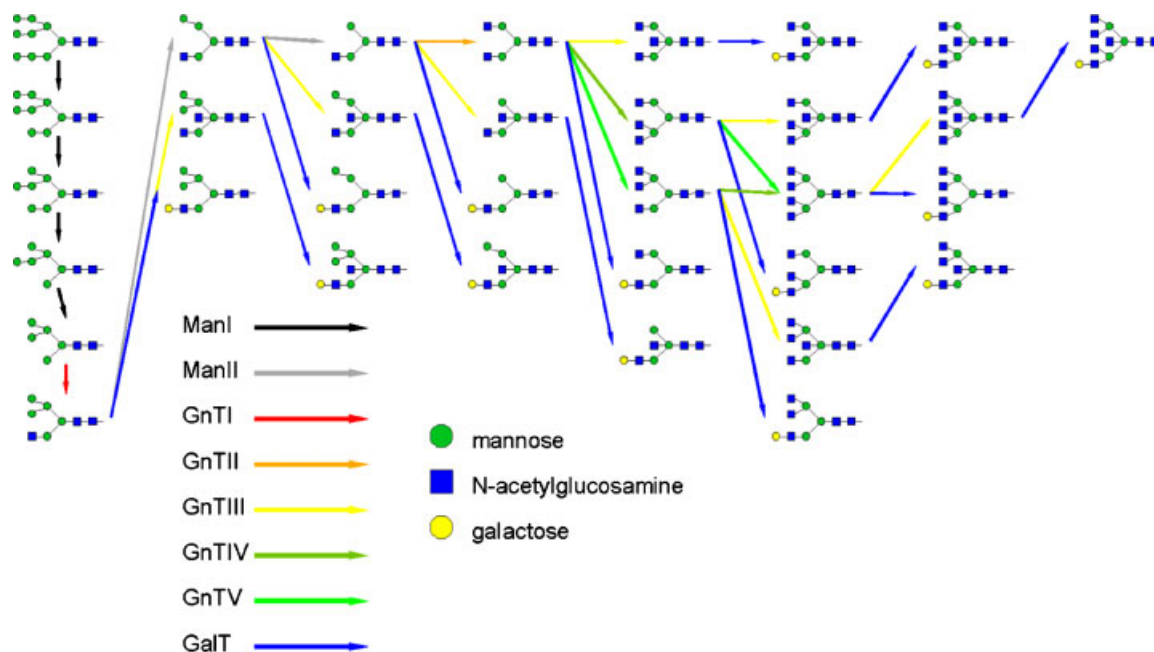


Figure 3. UB1997 reaction network (33 structures, 33 reactions). [Color figure can be seen in the online version of this article, available at www.interscience.wiley.com.]

Note that a rule table such as Table IV provides a compact summary of the behavior of all enzymes in a reaction system. Further, it is a simple matter to write a computer program that takes the rule table and the initial structure and generates the complete compound list and reaction network. Once this is done, it is straightforward to extend the model by the addition of more enzymes or alternatively to modify the rules for one or more enzymes and examine the effect on the overall network. For example, Table IV generates the network shown in Figure 3.

The first logical extension to the UB1997 model is to follow the individual isomers generated by galactosylation of the GlcNAc groups on the ends of the various antennae. This is done by adding three more rules for the GalT enzyme, shown as rules 13–15 in Table V.

Just the addition of the three additional galactosylation rules increases the number of structures from 33 to 89 and the number of reactions from 33 to 145.

In reviewing the assumptions of the UB1997 model it was found that a few were oversimplifications of the actual

Table V. Rules for current model (7,565 structures, 22,871 reactions). Part of the resulting reaction network is shown in Figure 5.

Index	Enzyme	Cosubstrate	Coproduct	Substrate rule	Man	Fuc	Gnb	Br1	Br2	Br3	Br4	Gal	Sla
1	ManI	Water	Mannose	Man > 5	−1								
2	ManII	Water	Mannose	Man > 3 & Br4 = 1 & Gnb = 0	−1								
3	FucT	GDP-Fuc	GDP	Fuc = 0 & Br4 > 0 & Gnb = 0 & Gal = 0		1							
4	GnTI	UDP-GlcNAc	UDP	Br4 = 0 & Man = 5							1		
5	GnTII	UDP-GlcNAc	UDP	Br2 = 0 & Man = 3 & Br4 = 1 & Gnb = 0					1				
6	GnTIII	UDP-GlcNAc	UDP	Gnb = 0 & Br4 > 0 & Gal = 0			1						
7	GnTIV	UDP-GlcNAc	UDP	Br3 = 0 & Br4 = 1 & Gnb = 0						1			
8	GnTV	UDP-GlcNAc	UDP	Br1 = 0 & Br2 = 1 & Gnb = 0				1					
9	GnTE	UDP-GlcNAc	UDP	Br1 = 2				2					
10	GnTE	UDP-GlcNAc	UDP	Br2 = 2					2				
11	GnTE	UDP-GlcNAc	UDP	Br3 = 2						2			
12	GnTE	UDP-GlcNAc	UDP	Br4 = 2							2		
13	GalT	UDP-Gal	UDP	Br1 = 1 or Br1 = 4				1				1	
14	GalT	UDP-Gal	UDP	Br2 = 1 or Br2 = 4					1			1	
15	GalT	UDP-Gal	UDP	Br3 = 1 or Br3 = 4						1		1	
16	GalT	UDP-Gal	UDP	Br4 = 1 or Br4 = 4							1	1	
17	SiaT	CMP-NeuAc	CMP	Br1 = 2 or Br1 = 5				1					1
18	SiaT	CMP-NeuAc	CMP	Br2 = 2 or Br2 = 5					1				1
19	SiaT	CMP-NeuAc	CMP	Br3 = 2 or Br3 = 5						1			1
20	SiaT	CMP-NeuAc	CMP	Br4 = 2 or Br4 = 5							1		1

glycosylation process. The authors assumed that GnTIV cannot operate unless GnTII has already added a GlcNAc to branch 2. Instead, GnTIV can operate once GnTI has added a GlcNAc to branch 4, thus allowing an additional branch (Br3) on the structures even without a GlcNAc on branch 2 (Schachter et al., 1989). These reactions are included in Figure 2 by eliminating the requirement for $Br2 > 0$ from rule 6 in Table IV. Also, the same reference states that GnTIV can still operate if branch 1 or branch 2 is galactosylated, albeit at a lower rate than if the branches are not galactosylated. (The lower rate is accommodated through adjustments rules, as explained in the “Model parameters” section below.) GnTIV will not operate if branch 3 is galactosylated. To express these facts the requirements “ $Br4 > 0$ & $Gal = 0$ ” in rule 6 should be replaced with “ $Br4 = 1$.” Also, a similar justification can be made for modifying the characteristics of GnTV enzyme activity. To accommodate this “ $Br2 > 0$ & $Gal = 0$ ” is replaced in rule 7 with “ $Br2 = 1$.” The changes are shown in Table V. The reaction network for this expanded version of the UB1997 model consists of 113 structures and 189 reactions and is shown in Figure 4.

Next rules are added for fucosylation by FucT, branch extension by GnTE, and sialylation by SiaT as shown in Table V. The rule for fucosylation ($Fuc = 0$ & $Br4 > 0$ & $Gnb = 0$ & $Gal = 0$) is based on the requirements that the structure contain at least one branch and that it not contain a

bisecting GlcNAc or a galactose group (Schachter et al., 1989). The rules for GnTE, GalT, and SiaT are straightforward, only requiring the appropriate acceptor group on the end of any branch. For GnTE the presence of a “2” in the column for the branch being extended indicates that the extension value increases from two to four when the enzyme acts. The rules of Table V generate 7,565 structures and 22,871 reactions. The first few generations of the reaction network are illustrated in Figure 5 but the complete network would be unwieldy to project graphically. In contrast, the representation in Table V contains the information for the complete reaction profile and is quite simple. Note that rules 9–12 for GnTE result in only one repeat *N*-acetylglucosamine group ($Gal\beta 1-3GlcNAc$) per branch. For situations where more repeats occur the rules can easily be modified to accommodate extra groups.

MATHEMATICAL MODEL

Transport Equations

It is assumed that the Golgi compartments of interest can be modeled as well-mixed reactors with bulk flow of the contents from each compartment to the next compartment in line while the enzymes remain fixed in the compartments. At steady state, the concentrations of the various structures

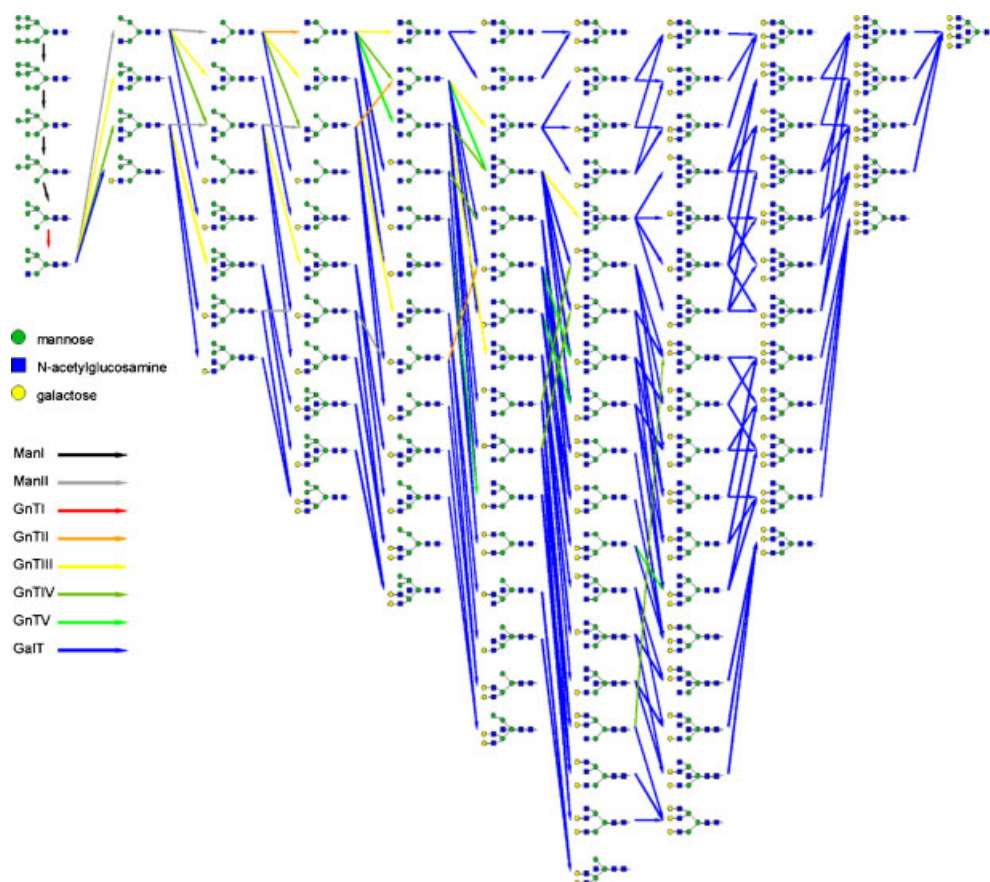


Figure 4. Extended UB reaction network (113 structures, 189 reactions). [Color figure can be seen in the online version of this article, available at www.interscience.wiley.com.]



Figure 5. Initial stages of full model (7,565 structures, 22,871 reactions). [Color figure can be seen in the online version of this article, available at www.interscience.wiley.com.]

satisfy the balance equations

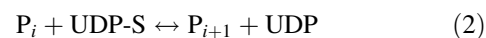
$$c_{ij} = c_{i,j-1} + \tau_j r_{ij} \quad (1)$$

where c_{ij} is the concentration of structure i in compartment j , τ_j is the residence time of compartment j , and r_{ij} is the net rate of production of structure i per unit volume of the compartment due to all the biochemical reactions that occur. Since the reactions considered do not create or destroy glycans the total concentration of N -glycans, c_{tot} , is the same in each compartment and is given by $p\tau_1/v_1$, where p is the total production rate of glycans and v_1 is the volume of compartment 1. In the balance equations for the first

compartment the concentration c_{i0} is given by the fraction of total glycans that initially have structure i multiplied by c_{tot} .

Reaction Kinetics

Consider the glycosylation of a polysaccharide P_i with a monosaccharide S catalyzed by an enzyme E . Assume the donor cosubstrate is UDP- S . The overall reaction is



As shown in detail in the Appendix, assuming that the product P_{i+1} competes for the same enzyme site as the substrate P_i , that the donor cosubstrate UDP- S occupies a second site on the enzyme, and that the reaction is reversible, the Michaelis–Menten kinetic equation takes the form

$$r = \frac{k_f[E_t]([[\text{UDP-S}][P_i] - 1/K'_{eq}[P_{i+1}][\text{UDP}]])}{K_{mi}(K_{md} + [\text{UDP-S}])(1 + \sum_j [P_j]/K_{mj})} \quad (3)$$

$$K'_{eq} = \frac{k_f K_{m,i+1}}{k_r K_{md} K_{mi}} = \frac{[P_{i+1}]^*[\text{UDP}]^*}{[P_i]^*[\text{UDP-S}]^*}$$

Here k_f and k_r are the forward and reverse rate coefficients, K_{mi} and K_{md} are the dissociation constants for the substrate and donor cosubstrate, and K'_{eq} is the apparent equilibrium constant for the overall reaction. The symbols $[\]^*$ in (3) denote equilibrium concentrations. The summation in the denominator is taken over all the substrates that compete for the same enzyme.

Equation (3) also assumes that the product nucleotide, UDP in this case, does not compete for the donor site. While inhibition of GlcNAc transferase by UDP has been observed (Nishikawa et al., 1988) it is not clear if this is due to competition for the enzyme site or reversibility of the reaction. In any event in the Golgi compartments the UDP is rapidly hydrolyzed to UMP, so its concentration is quite low. For the same reason the reverse reactions are neglected in the remainder of the paper. The reversible equation is included in the model, however, for possible use in analyzing in vitro data under reversible conditions.

Numerical Methods

The model equations are nonlinear algebraic equations which are solved for the concentrations of each of the structures in each of the four Golgi compartments. The case with 7,565 structures has 30,260 equations and unknowns. These are solved using a constrained Newton-Raphson method with the Harwell MA28 sparse linear solver (HSL, 2002). The efficiency of a sparse linear solver for large numbers of variables depends on the problem Jacobian being sparse. The Michaelis–Menten denominator terms in Equation (3) involve a large number of species that compete for each of the enzymes. This could make the Jacobian matrix rather dense. To avoid this the denominator terms for each enzyme are formulated as separate variables with equations added to specify how the denominators are calculated. This confines the equations with large numbers of variables to only one for

each enzyme. Analytical derivatives were used. While each compartment could be solved separately in sequence to give four subproblems, each one fourth the size, this was not found to be necessary.

In addition to solving the model for a given set of model parameters, provision was also made to adjust parameters to match a given set of data. This was done using the Marquardt–Levenberg method with numerically estimated derivatives (Marquardt, 1963). The same method was used for optimizing model parameters to achieve a desired distribution of glycan structures.

The Marquardt–Levenberg method is typical of nonlinear optimization algorithms in that it makes use of a sequence of local linear approximations to the nonlinear model to converge to a solution that is a local optimum. Except in special cases there is no way to determine whether the nonlinear problem possesses an even better solution far removed from this point. Experience in using this method shows, however, that if a reasonably good fit is obtained with the local optimum it is also a global optimum for the parameter estimation problem.

All required calculations could be done on a laptop computer.

MODEL PARAMETERS

Based on a thorough review of the literature, Umana and Bailey (1997) estimated values for the various parameters in their model. Most of these are used here and are supplemented with further estimates for the additional parameters required by the current model.

Protein Transport

In the UB1997 model the protein transport rate was specified as a value of k_G , the kinetic constant for protein transport in the Golgi. The value of k_G is just the reciprocal of the Golgi compartment residence time τ used here, so their value of 0.18/min becomes $\tau = 5.56$ min for each of the four compartments. This is very similar to the value of 5 min per compartment used in a model for the sialylation step of glycosylation developed by Monica et al. (1997).

Compartment Volume and Total Glycan Concentration

The volume of the Golgi apparatus is required to estimate the concentration of glycans in each compartment. The UB model assumes a total volume of $10 \mu\text{m}^3$ for the four compartments divided equally between them. Assuming glycan productivities in the range of 100–1,500 pmol/ 10^6 cells 24 h (as in UB1997), the corresponding c_{tot} range ($p\tau_1/v_1$) is 150–2,300 μM .

Spatial Distribution of Enzymes

The enzyme distribution in each compartment are based on an immunoelectron microscopy study of HeLa cells

(Rabouille et al., 1995) as noted in the UB1997 model. Fractional distributions for ManI, ManII, and GnTI–GnTV are 0.15, 0.45, 0.30, and 0.10, respectively in the four sequential Golgi compartments of the model. For GalT and SiaT, the enzymes are distributed in the following ratio: 0, 0.05, 0.20, and 0.75, in the four compartments. No enzyme distribution data was available for GnTE. This enzyme can only act after GalT has acted, therefore there was no accumulation assumed in compartment 1. Similarly, a large amount of GnTE in compartment 4 would limit sialylation. The distribution of GnTE is therefore assumed to be 0, 0.60, 0.30, 0.10.

Kinetic Parameters

Umana and Bailey (1997) estimated reaction rate parameters for the 33 reactions in their model based on a review of the literature. K_m values for the various enzymes were found for a variety of species. These were combined with V_m values measured for CHO cell lysates. They were able to derive rate parameters for the individual galactosylation reactions of the various terminal residues of a given structure from literature data on various substrates. These authors then lumped all galactosylation reactions for a given structure into a single reaction and calculated K_m values for the lumped reactions by taking the reciprocal of the sum of $1/K_m$ for each individual branch of the structure. They also made adjustments for other structural features based on literature information. Their results have here been translated into a set of rule-based adjustments. The parameter values and adjustments that generate the 33 sets of kinetic parameters for the UB1997 model are shown in Table VI. The adjustment factors in Table VI are multiplied by the corresponding kinetic parameters when the adjustment rule is satisfied. The V_m values in Table VI, expressed as pmol/ 10^6 cells-h, were derived in UB1997 from original values expressed as pmol/mg protein-h assuming that 1 mg of total protein results from 27.6×10^6 cells.

The total cell V_m values correspond to the product $k_f[E_t]v_i$ in Equation 3 of the current model. The k_f (turnover numbers) have been calculated from independently measured values for intrinsic enzyme activities shown in Table VII. The turnover number in min^{-1} is just the product of intrinsic activity in $\mu\text{mol}/\text{mg}$ enzyme-min and enzyme molecular weight, in kDa. The enzyme concentrations $[E_t]$ were then calculated, using a single compartment volume v_i of $2.5 \mu\text{m}^3$, in order to match the total cell V_m data from UB1997 as shown in Table VIII. These calculations assume all of the enzyme in the cell is concentrated into a single compartment. The concentration in each compartment is then obtained by multiplying by the enzyme distribution fractions given above. Since no CHO total cell V_m values were found for FucT, GnTE, or SiaT the $[E_t]$ values were initially assumed to be 1 μM , which is within the range typical for other enzymes in Table VIII.

While it was assumed in the UB1997 model that the enzymes are always saturated with donor cosubstrate, there

Table VI. Kinetic parameters for UB1997 model (V_m in pmol/10⁶ cells h, K_m in μ M).

Index	Enzyme	Substrate rule	V_m	K_m	Adjustment factors			
					Index	Rule	V_m	K_m
1	ManI	Man > 5	300	100	2	Man = 4	1	0.50
2	ManII	Man > 3 & Br4 = 1 & Gnb = 0	300	200	7	Br3 > 0	1	0.69
3	GnTI	Man = 5 & Br4 = 0	450	260	5	Br2 > 0	1	0.05
4	GnTII	Man = 3 & Br4 = 1 & Br2 = 0 & Gnb = 0	140	190	8	Gnb > 0 & Br2 > 0	1	3.62
5	GnTIII	Gnb = 0 & Br4 = 1 & Gal = 0	100	4,000	8	Br2 > 0	1	0.03
6	GnTIV	Br4 = 1 & Br2 = 1 & Br3 = 0 & Gnb = 0 & Gal = 0	10	3,400	8	Br1 > 0	1	0.54
7	GnTV	Br2 = 1 & Br4 = 1 & Br1 = 0 & Gnb = 0 & Gal = 0	10	130	8	Br3 > 0	1	0.38
8	GalT	Br4 = 1	580	4,000	8	Br1 > 0 & Br3 > 0	1	1.49

were K_m values for the donor reported in the same literature references. These additional values (as K_{md}) are shown in Table IX together with data for additional enzymes not present in the UB model but included in the current expanded version. K_{md} values were not found for GalT, so these will be assumed to be zero, which is the same as assuming saturation of the enzymes with donor.

As mentioned above GnTIV can act on more substrates than assumed in the UB1997 model. It can act before GnTII has acted to add branch 2 (Gn β 12M α 16) but with only 20% of the activity it would have after this event. It can also act if either branch 1 or branch 2 is galactosylated but with only 66% of the activity prior to this galactosylation (Schachter et al., 1989). Also, once GnTV has acted the K_m value for GnTIV is reduced by a factor of 0.178 (Oguri et al., 1997). Adjustment rules were added to cover these effects. Relative values for GnTE acting on various substrates were given by Schachter et al. (1989). Extension of branch 1 or branch 2 via GnTE is 10 times faster than extension of branch 3 or branch 4. When both branch 1 and branch 2 are present extension is four times faster than when only one is present. These observations were used to develop adjustment rules for GnTE. Also, in analyzing data on sialic acid addition patterns, discussed later, it was found that once two sialic acid residues are added the subsequent additions become

slower, so an adjustment for this was added. All the assumed values and adjustment rules for the current model are shown in Table X. Of course the enzyme concentrations are strongly dependent on species, cell line or tissue, and the kinetic parameters depend on configuration of the protein backbone, even varying for different glycosylation sites on the same protein (Kawasaki et al., 2000). Because of this the values in Table X may vary significantly in different situations.

Donor Concentrations

The abundances of UDP-HexNAc (mostly UDP-Gn), UDP-Hexose (mostly UDP-Gal) and CMP-SA were measured in CHO cells as 0.24, 0.10, and 0.06 fmol/cell, respectively (Baker et al., 2001). The volume of the cytoplasm for BHK cells has been estimated at 1,050 μ m³/cell (Griffiths et al., 1989). Assuming that CHO cells are roughly the same size as BHK cells, these levels would translate to concentrations of 230, 95, and 60 μ M, respectively, averaged over the cytoplasm. Due to active transport into the Golgi, the concentrations of these species inside the compartments can be higher by a factor of 20–100 (Perez and Hirschberg, 1985). As an initial estimate the donor concentrations are assumed to be 40 times these levels, or 9,200, 3,800, and 2,400 μ M, respectively.

Table VII. Measured intrinsic enzyme activities and corresponding turnover numbers.

Enzyme	Intrinsic V_m μ mol/mg-min	MW (kDa)	k_f , min ⁻¹	Source	Substrate	Reference
ManI		230		Rat liver		Tulsiani and Touster (1989)
ManII	6.75	285	1,924	Rat liver	4-nitrophenyl mannoside	Moremen et al. (1991)
FucT	4.37	58	253	Porcine brain	GlcNAc-GlcNAc-Asn-PABA	Uozumi et al. (1996)
GnTI	19.8	50	990	Rat liver	M5	Nishikawa et al. (1988)
GnTII	27.5	48	1,320	Rat liver	Man α 16(GlcNAc β 12Man α 13)RPA	Schachter et al. (1989)
GnTIII	5.52	110	607	Rat kidney	GlcNAc β 12Man α 16(GlcNAc β 12Man α 13)RPA	Nishikawa et al. (1992)
GnTIV	3.23	58	187	Bovine intestine	GlcNAc β 12Man α 16(GlcNAc β 12Man α 13)RPA	Oguri et al. (1997)
GnTV	18.80	75	1,410	Rat kidney	GlcNAc β 12Man α 16Man β -R	Shoreibah et al. (1992)
GnTE	0.27	90	25	Calf serum	GalGlcNAc	Kawashima et al. (1993)
GalT	19.8	440	8,712	Rat liver	GlcNAc	Bendiak et al. (1993)
SiaT	10.3	47	484	Porcine submax. gland	Antifreeze glycoprotein	Sadler et al. (1979)

Table VIII. Assumed enzyme concentrations (based on single compartment volume of 2.5 μm^3).

	k_f, min^{-1}	$V_m(\text{cell})$ $\text{pmol}/10^6$ cells-hr^*	$[E_i], \mu\text{M}$
ManI	1,924	300	1.04
ManII	1,924	300	1.04
FucT	253		1.00
GnTI	990	450	3.03
GnTII	1,320	140	0.71
GnTIII	607	100	1.10
GnTIV	187	10	0.36
GnTV	1,410	10	0.05
GnTE	25		1.00
GalT	8,712	580	0.44
SiaT	484		1.00

* V_m values from UB1997.

COMPARISON WITH UB1997 MODEL

The UB1997 model has been modified stepwise in order to see how the results would be affected by selective modifications. For the base case bisecting enzyme GnTIII concentration was set to zero, corresponding to the wild type CHO cells, c_{tot} was set to 800 μM , corresponding to a protein productivity of 518 $\text{pmol}/10^6$ cells in 24 h, and the number of glycans per protein molecule was fixed at one. The simulation results for this case are shown as fractions of total glycan in the first row of Table XI. For this table, the glycan structures have been lumped into seven categories: high mannose (HighM), which are the structures M_9 – M_5 , hybrid, which contain 4–5 mannose groups and at least one added Gn group, and one through four antennae (A1–A4) attached to a three mannose core. A3 and A3' refer to the triantennary structures with the third antenna on the $\text{M}\alpha 13$ branch and the $\text{M}\alpha 16$ branch, respectively.

The second row of Table XI shows the effect of following all the galactosylated species individually rather than considering them as lumped together. This change has little effect on the resulting distribution of the major classes of structures, as previously asserted by those authors.

The third row shows the effect of modifying the rules for GnTIV and GnTV as discussed above. This modification significantly increases the fraction of A3 and A4 structures. As discussed in UB1997, the first row of Table XI is much lower in tri- and tetraantennary structures than observed for glycosylation sites that carry complex N-glycans, so it is believed this change is appropriate for many systems.

The fourth row shows the effect of using nonzero K_m values for the donor cosubstrates, with the donor concentrations set at 100 times the concentration based on cytoplasm volume. This gives a value for UDP-Gn concentration in the compartments of 23 mM. Even at this higher donor–nucleotide concentration, there are significant effects from including the effect of cosubstrate concentration on the results, indicating that all the enzymes are not saturated as previously assumed. This finding is also borne out by several experimental studies that indicated an effect of the donor nucleotide concentration on glycosylation profile (Baker et al., 2001; Sasai and Ikeda, 2002). The fifth row of Table XI shows that the effect is even larger if the donors are reduced to 40 times the estimated bulk values. The effect of donor concentration can be seen by comparing rows 3, 4, and 5 of Table XI. In row three the enzymes were assumed to be saturated with donors, so that it corresponds to infinite donor concentration. As the donor concentrations are reduced in rows 4 and 5 it can be seen that the distribution shifts from A3, A3', and A4 species to A1 and A2 species. This is clearly due to the high K_{md} values for GnTIV and GnTV, which initiate branch 3 and branch 1, respectively.

In addition to these changes the current model incorporates sialylation and extension of the antennae. As expected this

Table IX. Apparent dissociation constants.

Enzyme	$K_m (\mu\text{M})$	$K_{md} (\mu\text{M})$	Source	Acceptor	Reference
ManI	100		Rat liver	M8	Tabas and Kornfeld (1979)
FucT	25	46	Porcine brain	GlcNac β 14GlcNac-Asn-PABA	Uozumi et al. (1996)
GnTI	260	170	CHO cells	M5	Chaney and Stanley (1986)
GnTII	190	960	Rat liver	Man α 16(Man α 13)Man β 14GlcNac β 14GlcNac	Schachter et al. (1989)
GnTIII	190	3,100	Rat	Man α 16(Man α 13)Man β 14GlcNac β 14GlcNac-PA	Taniguchi et al. (1989)
GnTIII	230	1,100	Hen oviduct	Man α 16(Man α 13)Man β 14GlcNac β 14GlcNac	Schachter et al. (1989)
GnTIV	3,400	8,300	Rat	Man α 16(Man α 13)Man β 14GlcNac β 14GlcNac	Taniguchi et al. (1989)
GnTV	130	3,500	Human lung	Man α 16(Man α 13)Man β 14GlcNac β 14GlcNac	Gu et al. (1993)
GnTV	150	4,000–4,600	Human-rec	Man α 16(Man α 13)Man β 14GlcNac β 14GlcNac-PA	Sasai and Ikeda (2002)
GnTV	270	7,600	Rat	Man α 16(Man α 13)Man β 14GlcNac β 14GlcNac-PA	Taniguchi et al. (1989)
GnTE	700		Novikoff tumor	Gal β 14GlcNac β 16(Gal β 14GlcNac β 12)Man	Schachter et al. (1989)
GnTE	3,000		Novikoff tumor	Gal β 14GlcNac β 14(Gal β 14GlcNac β 12)Man	Schachter et al. (1989)
GnTE	9,000	55	Human	Gal β 14GlcNac	Piller and Cartron (1983)
GnTE	600		Rat	asialo- α 1-acid glycoprotein	van den Eijnden et al. (1988)
GalT	130		Rat liver	Man α 16(Man α 13)Man β 14GlcNac β 14GlcNac	Paquet et al. (1984)
SiaT	260	57	Rat liver	asialo α 1 acid glycoprotein	Williams et al. (1995)

Table X. Assumed kinetic parameters and adjustment rules for current model.

Index	Enzyme	Substrate rule	k_f	K_m	K_{md}	Adjustment factors				
						Index	Rule	k_f	K_m	K_{md}
1	ManI	Man > 5	1,924	100	0					
2	ManII	Man > 3 & Br4 = 1 & Gnb = 0	1,924	200	0	2	Man = 4	1	0.5	1
3	FucT	Fuc = 0 & Br4 > 0 & Gnb = 0 & Gal = 0	253	25	46	6	Br2 > 0	1	0.048	1
4	GnTI	Man = 5 & Br4 = 0	990	260	170	7	Br2 = 0	1	5	1
5	GnTII	Man = 3 & Br4 = 1 & Br2 = 0 & Gnb = 0	1,320	190	960	7	Br2 > 1 or Br1 > 1	1	1.5	1
6	GnTIII	Gnb = 0 & Br4 > 0 & Gal = 0	607	4,000	3,100	7	Br1 > 0	1	0.178	1
7	GnTIV	Br3 = 0 & Br4 = 1 & Gnb = 0	187	3,400	8,300	8	Br3 > 0	1	0.692	1
8	GnTV	Br1 = 0 & Br2 = 1 & Gnb = 0	1,410	130	3,500	9	Br2 = 0	1	4	1
9	GnTE	Br1 = 2	25	700	55	10	Br1 = 0	1	4	1
10	GnTE	Br2 = 2	25	700	55	11	Br4 = 0	1	4	1
11	GnTE	Br3 = 2	25	7,000	55	12	Br3 = 0	1	4	1
12	GnTE	Br4 = 2	25	7,000	55	13	Gnb > 0 & Br2 > 0	1	3.62	1
13	GalT	Br1 = 1 or Br1 = 4	8,712	150	0	14	Gnb > 0 & Br2 > 0	1	3.62	1
14	GalT	Br2 = 1 or Br2 = 4	8,712	135	0	15	Gnb > 0 & Br2 > 0	1	3.62	1
15	GalT	Br3 = 1 or Br3 = 4	8,712	80	0	16	Gnb > 0 & Br2 > 0	1	3.62	1
16	GalT	Br4 = 1 or Br4 = 4	8,712	4,000	0	17	Sia > 1	1	5	1
17	SiaT	Br1 = 2 or Br1 = 5	484	260	57	18	Sia > 1	1	5	1
18	SiaT	Br2 = 2 or Br2 = 5	484	260	57	19	Sia > 1	1	5	1
19	SiaT	Br3 = 2 or Br3 = 5	484	260	57	20	Sia > 1	1	5	1
20	SiaT	Br4 = 2 or Br4 = 5	484	260	57					

has little effect on the gross structural types, as shown in the row labeled “Current model.” In contrast, when the model parameters are adjusted to match actual experimental glycoform structures as discussed below, the results are quite different, as seen in the row labeled “Current model (adj.).”

COMPARISON WITH EXPERIMENTAL DATA

Inoe et al. (1999) measured the structures of *N*-glycans on recombinant human thrombopoietin (TPO) produced in CHO cells. The six possible *N*-glycosylation sites of TPO were mostly glycosylated, with an average of 5.4 *N*-glycans per molecule. Because of this the structures reported are a composite of those from the six sites. The glycans were mostly sialylated, with one to four sialic acid groups per glycan. Essentially all the structures were fucosylated. The distribution of sialic acid groups is shown in Table XII and the detailed structures of desialylated *N*-glycans in Table XIII. These are the structures that were obtained after the sialic acid groups were removed using sialidase from *Arthrobacter ureafaciens*.

Each of the desialylated structures in Table XIII can result from a number of sialylated structures following removal of

the sialic acid residues. There are four possible sialylated structures for the biantennary structures, eight for each triantennary and 16 for each tetraantennary desialylated structure. As a result, the ten desialylated structures shown can result from 108 different sialylated ones.

Because the results are for wild-type CHO cells, which contain no GnTIII activity, this enzyme was dropped from the model. Since essentially all the species were fucosylated, and since there is little effect of fucosylation on the activity of the other enzymes, FucT was also dropped. The authors did not observe any *N*-acetylglucosamine repeats on branches 3 or 4 so the rules leading to these structures were also dropped. These changes reduced the network to 546 structures and 1,484 reactions which speeded the calculations for parameter fitting. The final results were checked with the reactions of FucT and all the *N*-acetylglucosamine repeats included, showing that the full model agrees with the measured high level of fucosylation and very low extension of branches 3 and 4. Changes in the calculated distributions were negligible for this case.

Values calculated from the model are also shown in Tables XII and XIII. No data was given by the authors on protein production rate, but another study reported a

Table XI. Effect of changes to the UB1997 model.

	HighM	Hybrid	A1	A2	A3	A3'	A4
UB1997 model	0.001	0.022	0.040	0.797	0.008	0.130	0.002
Galactosylation details	0.001	0.020	0.037	0.790	0.009	0.140	0.003
Rule change	0.001	0.020	0.037	0.769	0.027	0.118	0.029
Donor effects (100×)	0.001	0.020	0.039	0.792	0.021	0.108	0.019
Donor effects (40×)	0.001	0.020	0.042	0.815	0.015	0.094	0.012
Current model	0.001	0.023	0.044	0.812	0.015	0.093	0.012
Current model (adj.)	0.000	0.004	0.007	0.321	0.039	0.274	0.354

Table XII. Distribution of sialic acid groups on *N*-glycans of recombinant human TPO (Inoue et al., 1999).

Number of sialic acid groups	0	1	2	3	4
Percent of <i>N</i> -glycans (measured)	1	12	47	27	12
Calculated (unadjusted)	3	15	68	13	1
Calculated (adjusted)	3	10	48	26	13

production rate of recombinant human TPO from CHO cells of 3.5 $\mu\text{g}/10^6$ cells-day (Kasubaska et al., 2000). Using a molecular weight of 85 kDa and 6 glycans per molecule gives a c_{tot} of 250 μM . No information was given on total glycoproteins. For comparison with the model, a value of 500 μM has been used to take into account other glycoproteins simultaneously produced. The unadjusted model results were calculated assuming the enzyme concentrations of Table XIII and the parameter values of Table X, except without the adjustment rule for SiaT, which was added as part of the adjustment process. This rule decreases the sialylation rate by a factor of 5 once there are already two sialic acid groups on the structure. Without this adjustment a good fit to the sialic acid distribution could not be obtained. The lower rate of sialylation for bisialylated structures could be related to electrostatic repulsion of the negatively charged sialic acid groups. For the adjusted model the sialic acid rule was added and the enzyme concentrations were allowed to vary from their assumed values. This was done using the parameter estimation method mentioned above. The FucT concentration was adjusted to give a concentration of unfucosylated

Table XIII. Distribution of desialylated structures for recombinant human TPO (Inoue et al., 1999).

Code	Structure	Measured amount	Calculated (unadjusted)	Calculated (adjusted)
310020220		27.9	77.8	27.4
310022230		5.9	1.6	4.2
310220230		16.2	11.8	17.6
310222240		22.6	1.6	22.5
310520240		3.4	0.8	3.4
310250240		4.0	0.8	3.5
310522250		5.2	0.1	6.7
310252250		5.7	0.1	6.8
310550250		2.7	0.1	1.0
310552260		5.2	0.0	2.9

structures of less than 1% as determined experimentally. The original and adjusted enzyme concentrations are listed in Table XIV.

A choice was made to adjust enzyme concentrations to match the data rather than other parameters, such as reaction rate coefficients or dissociation constants. This was somewhat arbitrary but seems reasonable based on the fact that the amounts of the various enzymes present in a cell can vary significantly for different cell lines, for different tissues of the same organism or even for the same cell under different conditions. The kinetic properties of the enzymes are somewhat less variable. The main issue, though, is how different choices of adjustable variables might affect model predictions. For example if rate coefficients had been chosen for adjustment instead of enzyme concentrations it would make no difference at all in model predictions, since these two quantities only occur in the model as a product of the two. On the other hand if dissociation constants were chosen then predictions of the effect of total glycan concentration would be different. This could only be resolved by simultaneously performing experiments at varying levels of total glycan concentration. If the results were different from the predictions we would then adjust concentrations and dissociation constants simultaneously to match the data.

Of the assumed enzyme concentrations only those of GnTI, GnTII, GnTV, and GalT were based on actual data for CHO cells. (Umana and Bailey made guesses for ManI, ManII, and GnTIV which were adopted here.) Of the values based on actual measurements the largest adjustment was made for GnTV. Since the assumed value was surprisingly low compared to the other enzymes the adjusted value seems reasonable. It is also possible that the assumed total compartment volume of 10 μm^3 is too low, which increases the total glycan concentration at any given productivity, thus increasing the inhibition of reaction rates and requiring higher enzyme content as indicated in the adjusted model. Indeed, measured volumes of Golgi compartment for BHK cells suggest that the volume could be higher (Griffiths et al., 1989). The final two structures in Table XIII are somewhat under predicted both in the assumed and adjusted model. These findings suggest that once one of the two branches on the $\alpha(1,6)$ mannose has been extended with an *N*-acetylglucosamine residue, the extension of the other branch is

Table XIV. Adjustments to assumed enzyme concentrations to match data.

	Assumed $[E_i]$, μM	Adjusted $[E_i]$, μM
ManI	1.04	1.78
ManII	1.04	1.32
FucT	1.00	2.50
GnTI	3.03	3.05
GnTII	0.71	1.29
GnTIV	0.36	3.62
GnTV	0.05	0.40
GnTE	1.00	3.47
GalT	0.44	0.66
SiaT	1.00	1.00

enhanced. If desired, an adjustment rule could be added to match this behavior.

USE OF MODEL FOR STRUCTURE OPTIMIZATION

One of the main features of this model will be to guide experimentation in order to modify the distribution of glycan structures on recombinant proteins for improved therapeutic efficacy.

Increased Protein Expression and Glycosylation

Various methods can be used to increase protein expression in cultured cells. This results in an increased rate of protein transport through the Golgi stack with a corresponding increase in glycan concentration. The results in Figures 6, 7, and 8 illustrate how this variable affects the final *N*-glycan structures. Shown in Figure 6 is the effect of total glycan concentration on the antennarity of the *N*-glycans. Here A0 refers to any structure with more than three mannose residues. While the effect at concentrations below about 1,000 μM is fairly minor, higher concentrations reduce the level of antennarity, decreasing A4 and A3' while increasing A1 and A2 structures. The effect on *N*-acetylglucosamine repeats is shown in Figure 7. In Figure 7, L1 and L1' represent a single repeat on branch 1 or branch 2, respectively, while L2 represents a repeat on both branches. All of these decrease with glycan concentration level, even at lower concentrations. Similarly, the degree of sialylation, where S0–S4 represent structures with zero through four sialic acid groups, decreases as shown in Figure 8

This behavior is intimately related to the K_m values of the various enzyme–substrate combinations. If the kinetics were linear, with reaction rate directly proportional to substrate concentration, there would be no effect of total concentration on product distribution, since the reactor residence times are fixed. In this case, however, the Michaelis–Menten denominator terms prevent the reaction rates from keeping up with

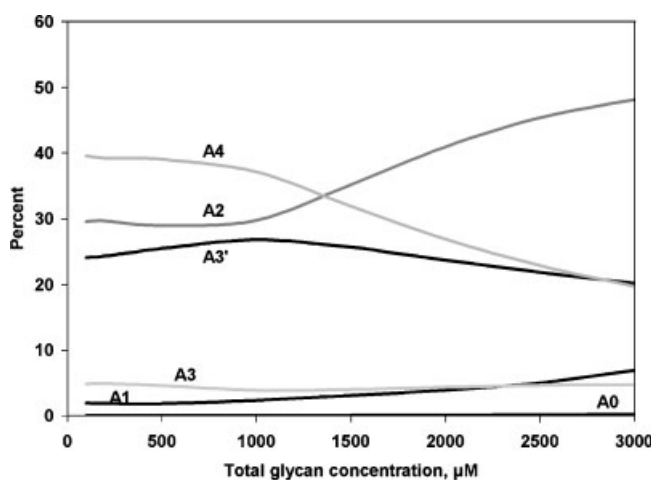


Figure 6. Effect of total *N*-glycan concentration (c_{tot}) on antennarity. *N*-glycan concentration is proportional to productivity. A total *N*-glycan concentration of 1,000 μM corresponds to a productivity of 650 pmol/10⁶ cells-day.

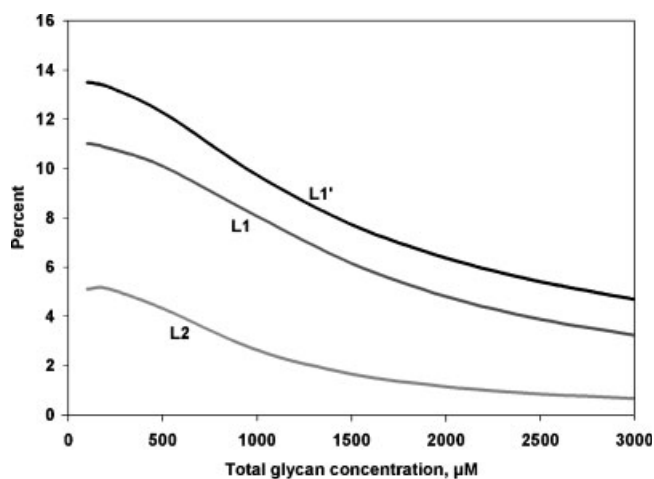


Figure 7. Effect of *N*-glycan concentration on repeat *N*-acetylglucosamine groups. L1: one repeat on branch 1. L1': one repeat on branch 2. L2: two repeats.

the increased concentrations of *N*-glycans. Reactions with lower K_m values saturate at lower concentrations. Since the enzymes GnTIV and GnTV, which create branches 1 and 3, have relatively high K_m values the enzymes do not saturate as quickly as GnTE or SiaT as concentration increases.

Suppose it is desired to increase protein productivity of a cell culture while preventing the glycan distribution from changing. Consider an example where a fourfold increase in protein productivity is achieved by adjusting culture conditions (adding sodium butyrate, for example). In the model, this corresponds to a fourfold increase in the parameter c_{tot} , from 500 to 2,000 μM . This shifts the distribution of glycan structures significantly as shown in Case 1 of Table XV. The subsequent cases of Table XV show how various changes in model parameters can reduce the deviation of the glycan distribution from the base case. The changes considered were in enzyme concentration, which could be accomplished by modifications that overexpress the genes that encode these enzymes, and in UDP-GlcNAc concentration, which can be increased by adding glucosamine and uridine to the culture medium (Baker et al., 2001). The changes were determined

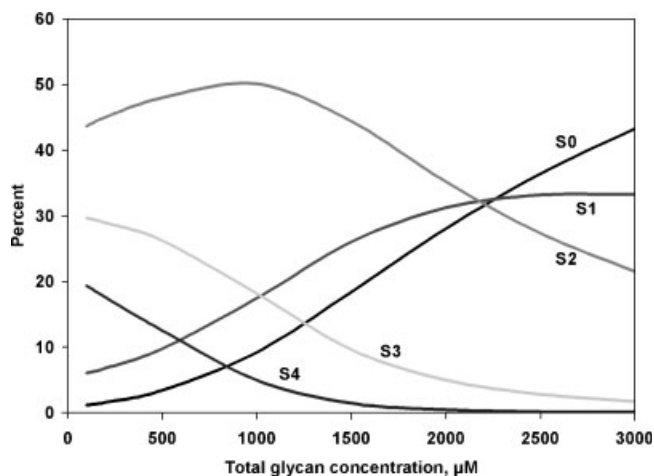


Figure 8. Effect of total *N*-glycan concentration on sialylation level.

Table XV. Adjustments to restore initial *N*-glycan distribution at four times higher productivity.

	Base	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6
ctot, μ M	500	2,000	2,000	2,000	2,000	2,000	2,000
ManI	1.00	1.00	1.00	1.00	5.62	1.00	1.00
ManII	1.00	1.00	1.00	1.00	1.00	1.00	1.00
GnTI	1.00	1.00	1.00	1.00	1.00	1.00	1.00
GnTII	1.00	1.00	1.00	1.00	1.00	1.00	1.00
GnTIV	1.00	1.00	1.00	1.00	1.00	1.00	1.00
GnTV	1.00	1.00	1.00	1.00	1.00	1.00	1.19
GnTE	1.00	1.00	1.00	3.20	2.93	2.82	2.72
GalT	1.00	1.00	1.00	1.00	1.00	1.00	1.00
SiaT	1.00	1.00	2.34	2.52	2.46	2.39	2.38
UDP-GlcNAc	1.00	1.00	1.00	1.00	1.00	2.95	1.88
Deviation	0.00	6.63	3.42	2.43	1.54	0.93	0.56
A0	0.02	0.09	0.09	0.09	0.01	0.09	0.09
A1	1.85	3.91	3.91	3.74	1.31	3.36	3.41
A2	28.98	40.97	41.00	40.40	36.47	31.72	29.23
A3	4.61	4.39	4.39	4.34	4.43	4.82	3.57
A3'	25.48	23.72	23.71	23.97	26.31	20.68	25.14
A4	39.08	26.93	26.91	27.47	31.47	39.33	38.57
L1'	12.28	6.37	5.82	13.04	12.96	12.39	12.26
L1	10.10	4.80	4.46	8.93	9.35	9.51	9.78
L2	4.32	1.14	0.89	5.22	4.64	4.60	4.48
S0	3.44	28.08	4.69	3.97	3.91	4.62	4.64
S1	9.81	31.23	12.89	11.86	10.38	11.83	11.76
S2	47.98	35.19	50.84	49.83	49.33	46.18	45.52
S3	26.23	5.01	21.75	22.93	24.54	23.81	24.96
S4	12.54	0.49	9.83	11.42	11.84	13.56	13.13

Deviation of the *N*-glycan distribution from the base is shown as RMS% change. Distributions are shown as mol% of total *N*-glycans.

by choosing one or more of these variables at a time and using the fitting algorithm to find the value of this variable or variables that best match to the base case distribution.

In Case 2 of Table XV the concentration of SiaT was increased by a factor of 2.34. This closely matched the level of sialylation, S0 through S4, of the base case but had little effect on branch extension of antennarity. In case 3, the level of GnTE was also increased. This matches both the sialylation level and the extension level, L1 through L2, of the base case but does not improve the antennarity. In Case 4, the best choice of a third enzyme to adjust turned out to be Man I. This is unexpected because there is very little of the high-mannose structures present in the product for any of the cases. It occurs because by speeding up the elimination of the M9–M6 structures acted upon by ManI the subsequent action of all the Gn transferases is promoted. The inclusion of higher levels of this enzyme improves the agreement of A1 through A4 with the base case. Case 5 shows that an even better match for antennarity can be obtained by adjusting the UDP-Gn level together with just the two enzymes, SiaT and GnTE. Case 6 shows that an almost perfect match of glycan distribution with the base case can be obtained by also adjusting GnTV to bring the amount of A3' into line.

Donor compounds other than UDP-Gn were not adjusted but it is clear that only the Gn transferases are significantly

affected by donor concentrations because of their relatively high K_{md} values in Table X. Indeed it has been shown experimentally that elevating the concentration of CMP-sialic acid has very little effect on degree of sialylation while UDP-Gn significantly enhances antennarity (Baker et al., 2001).

DISCUSSION

Using enzyme reaction rule tables provides an effective way of dealing with reaction networks generating highly variable structures such as glycoforms. Combining these with rate parameter adjustment rules makes it practical to construct models for such networks with reasonable numbers of parameters. It takes only a few minutes to make changes or additions to the rules and generate a new model. The current model includes branch extensions and modifications far beyond the previous UB1997 model. It also relaxes a number of simplifying assumptions of that work that have significant effects on glycoform structure.

All of the assumed parameter values in the current model are based on in vitro experiments with individual enzymes and soluble oligosaccharides. It is therefore remarkable that the combined model makes realistic predictions for in vivo results on complete glycoproteins with relatively minor adjustments. Of course a true test would involve measure-

ments and predictions of individual effects of enzyme expression level and donor concentrations combined with more detailed analyses of the structures.

The current version of the model leaves out a number of known enzymes that occur in various cell lines. For example, in addition to $\alpha(2,3)$ -linked sialic acid groups the terminal $\beta(1,4)$ -linked galactose of an antenna can accept $\alpha(2,6)$ -linked sialic acid, fucose, $\alpha(1,3)$ -linked galactose, or *N*-acetylgalactosamine. Of course, these can relatively easily be added as needed for a particular cell line. The simplified structural description in terms of a set number of configurations for each branch can be extended to cover more types of structures and still has a way to go before something more sophisticated will be needed. The current computer implementation is quite flexible, making addition of new enzymes, rules and structures simple. The solution methods used are efficient and can handle larger problems if needed.

The glycosylation process is highly variable. The structures depend on the specific protein being glycosylated and even on the particular glycosylation site on a protein. Different cell types of the same organism have different levels of the various enzymes involved. To make useful predictions of variable effects it will always be necessary to adjust the model parameters to some base case. As a rule, there will always be more parameters that can be adjusted than there are measurements to match. In this case it was chosen to adjust enzyme concentrations rather than K_m values to match the experimental distributions, since these are probably the most variable factors. If data on the effects of total concentration and more detailed structure information were available other parameters could be adjusted.

While studying the glycosylation pattern of the sites on a single protein we need to recognize that glycans on other proteins are competing for the same enzymes. This competition causes the glycosylation of the protein of interest to be coupled to that of all the others. As a first approximation this has been handled by setting the c_{tot} value using the total productivity of glycans for all proteins and sites processed in the Golgi stack. A more accurate approximation could be obtained by including two sets of rules and parameters that apply to two different sets of glycans, one set for a particular protein or individual site of interest and another set for everything else. This would require knowledge of the overall glycan structures for the total set of glycoproteins produced by the cell. However, since the specific protein considered here has a number of glycosylation sites (6) the average glycosylation pattern of all the glycoproteins produced by the cell may not be too different, in which case the first approximation would be accurate.

But the model is far more constrained than might be deduced from the large number of parameters involved. Because of its fundamental nature it automatically enforces stoichiometric constraints and correlations between fluxes (reactions) with common enzymes or substrates, so that all our knowledge of the underlying reaction network is included. These are far stronger constraints than the purely stoichiometric ones used in constraint-based modeling, for

example. Also, as model predictions are tested, more and more information about the particular system under study, including in vitro experiments far from in vivo conditions, can be incorporated through further parameter adjustments. The model may be thought of as a learning tool that can capture what is known about a particular system in a consistent way and predict the consequences of this knowledge. Because of the complex interactions of the various enzymes involved the model can predict effects that would be hard to guess intuitively, such as the increase in antennarity that results from increasing the activity of ManI shown above.

Furthermore, the model predictions could turn out to be quite useful as a guide to experimentation. Even more exciting is the idea of using the model to generate desirable (or eliminate undesirable) glycoform distributions. One could envision measuring the detailed glycan distributions for a desired human protein and a recombinant version of the same protein. The parameter values would then be adjusted to match the recombinant distribution, and an optimization calculation similar to that in the previous example would be done to predict the minimum variations in the cellular genome or growth medium to shift the recombinant distribution toward the human pattern. In vitro postprocessing with specific enzymes could also be considered. The indicated experiments would then be performed and agreement with the predictions evaluated. If the agreement is good the desired product is obtained. If not the model can be adjusted to match both the original and new result and a new optimization performed. This could conceivably result in a much faster route to the desired product glycoform distribution pattern. In cases where particular glycoforms are implicated in initiating diseases, model calculations may suggest strategies for blocking only those glycoforms from being produced, possibly inhibiting progress of the disease.

NOMENCLATURE

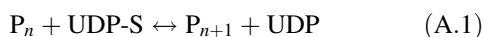
c_{ij}	concentration of structure i in compartment j , μM
c_{tot}	total concentration of N-glycans, μM
K_{eq}^i	apparent equilibrium constant
k_f	forward reaction rate coefficient (turnover number), min^{-1}
K_{md}	dissociation constant for donor cosubstrate, μM
K_{mi}	dissociation constant for substrate i , μM
k_r	reverse reaction rate coefficient (turnover number), min^{-1}
p	total production rate of glycans, $\mu\text{mol}/\text{min-cell}$
r	reaction rate, $\mu\text{mol}/\text{L-min}$
r_{ij}	net rate of production of structure i per unit volume in compartment j
v_j	volume of compartment j , μm^3
V_m	maximum reaction rate, $\text{Pmol}/10^6 \text{ cells-h}$, or intrinsic reaction rate, $\mu\text{mol}/\text{mg enzyme-min}$
τ_j	residence time of compartment j , min

APPENDIX

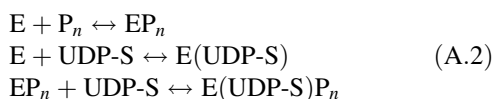
Kinetic Equations

Consider the reversible glycosylation of a polysaccharide P_n with a monosaccharide S catalyzed by an enzyme E .

Assuming UDP as the donor nucleotide, the overall reaction is



Let us assume that the polysaccharide P_n and the UDP-S equilibrate with separate sites on the enzyme E:



Then the concentrations of the various enzyme complexes satisfy

$$\begin{aligned} [EP_n] &= \frac{[E][P_n]}{K_{mn}} \\ [E(\text{UDP-S})] &= \frac{[E][\text{UDP-S}]}{K_{md}} \\ [E(\text{UDP-S})P_n] &= \frac{[E][\text{UDP-S}][P_n]}{K_{md}K_{mn}} \end{aligned} \quad (\text{A.3})$$

where K_{mn} and K_{md} are the dissociation constants for polysaccharide P_n and donor cosubstrate UDP-S, respectively.

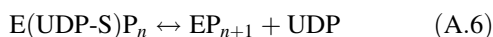
The total enzyme concentration, $[E_t]$, is then given by

$$\begin{aligned} [E_t] &= [E] + \frac{1}{K_{md}} [E][\text{UDP-S}] \\ &+ [E] \sum_j \frac{[P_j]}{K_{mj}} + \frac{1}{K_{md}} [E][\text{UDP-S}] \sum_j \frac{[P_j]}{K_{mj}} \end{aligned} \quad (\text{A.4})$$

Here the summation is over all the oligomers present that compete for the same enzyme site. Combining (A.3) and (A.4) and rearranging,

$$\begin{aligned} [EP_n] &= \frac{K_{md}[E_t][P_n]}{K_{mn}(K_{md} + [\text{UDP-S}]) \left(1 + \sum_j \frac{[P_j]}{K_{mj}} \right)} \\ [E(\text{UDP-S})P_n] &= \frac{[E_t][\text{UDP-S}][P_n]}{K_{mn}(K_{md} + [\text{UDP-S}]) \left(1 + \sum_j \frac{[P_j]}{K_{mj}} \right)} \end{aligned} \quad (\text{A.5})$$

The rate of the controlling reaction



is given by

$$r = k_f [E(\text{UDP-S})P_n] - k_r [EP_{n+1}][\text{UDP}] \quad (\text{A.7})$$

where k_f is the rate coefficient for the forward reaction (equal to the turnover number for the enzyme) and k_r is the rate coefficient for the reverse reaction. Combining (A.7) with (A.5) then gives the overall rate expression

$$\begin{aligned} r &= \frac{k_f [E_t] ([\text{UDP-S}][P_n] - 1/K'_{eq} [P_{n+1}][\text{UDP}])}{K_{mn}(K_{md} + [\text{UDP-S}]) \left(1 + \sum_j \frac{[P_j]}{K_{mj}} \right)} \\ K'_{eq} &= \frac{k_f K_{mn(n+1)}}{k_r K_{md} K_{mn}} = \frac{[P_{n+1}]^* [\text{UDP}]^*}{[P_n]^* [\text{UDP-S}]^*} \end{aligned} \quad (\text{A.8})$$

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